Effect of Iron & Its Supplementation on Athlete – A Review

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ABSTRACT

Iron insufficiency is common in sportspeople. Every kind of iron shortage needs to be treated since it can impair physical performance. Iron insufficiency in athletes is primarily caused by three mechanisms: increased iron demand, heightened iron loss, and iron absorption blockage by hepcidin bursts. Haemoglobin, haematocrit, mean cellular volume, mean cellular haemoglobin and serum ferritin levels are used as a baseline panel of blood tests to detect iron deficiency. Ferritin readings <15 mcg are indicative of empty, and values between 15 and 30 mcg/l are indicative of low iron stores in healthy male and female athletes over the age of 15. Consequently, cut-off of 30 mcg/l is suitable.

Cut-offs of 15 and 20 mcg/l, respectively, are advised for younger adolescents (12–15 years) and children (6–12 years). Athletes should achieve a ferritin level of 50 mcg/l before beginning altitude training, with the exception of adult elite sports, where there are higher iron needs. Iron deficiency is treated with dietary guidance, oral iron supplements, or, in certain situations, intravenous infusion. Intermittent oral replacement is beneficial for athletes whose ferritin levels are consistently low. It is crucial to monitor each athlete individually and to repeat the baseline blood tests mentioned above twice a year. Continuous daily oral iron consumption or supplementation in the event of normal or even elevated ferritin values does not make sense and may be harmful.

Keywords: Iron supplementation, performance, and athletic training

INTRODUCTION

Getting enough nutrients is crucial to performing at your best as an athlete. Iron is a necessary micronutrient in the processes that produce energy and is a part of what makes myoglobin and haemoglobin operate. It is thought that

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female athletes are more likely to have compromised iron status, which can result in iron deficiency (with or without anaemia) because of negative iron balance caused by inadequate dietary iron intake, menstruation, increased iron losses linked to hemolysis, perspiration, gastrointestinal bleeding, and acute inflammation brought on by exercise (Alaunyte et al., 2015).

Intramuscular or intravenous injections, oral supplements, dietary iron therapies, such as dietary counselling and advice, incorporating naturally iron-rich or fortified foods into the diet, and altering eating patterns, are all used to treat iron deficiency (Alaunyte et al., 2015). Oral iron supplements and injections, as conventional therapies, have been demonstrated to enhance athletes' iron status. However, they can also have unanticipated or unmonitored side effects, increasing the risk of iron overload and causing nausea, constipation, and abdominal discomfort. Therefore, it is recommended that diet be changed as the best way to guarantee sufficient iron intake, maintain iron status, and be the first line of defence against iron deficiency in female athletes (Alaunyte et al., 2015). Products supplemented with iron have been effectively utilised to raise serum ferritin and haemoglobin levels as well as lower the risk of iron insufficiency in the adult population at large. Dietary interventions to enhance iron status in female athletes have yielded inconsistent outcomes. Recent research has shown that including naturally iron-rich foods in one's diet will improve iron status in women who are fertile and recreational female runners (Alaunyte et al., 2015). The research examining the impact of dietary iron treatments on the iron status and supplementation for athletes will be the main focus of this review.

Dietary iron:

There are two types of dietary iron: hem and non-hem. All other forms of dietary iron are classified as non-haem iron, except for iron derived from animal sources, which is known as the haem form. Proteolytic enzymes in the lumen of the stomach and small intestine liberate haemoglobin, which contains haemoglobin or myoglobin (Alaunyte et al., 2015). Hem iron uptake can be as high as 40% since it does not require binding proteins. It should be remembered, nevertheless, that only roughly 10% of the iron in food comes via haems. Non-hem iron is typically found in ferric form and is bonded to other food ingredients. It needs to be converted to ferrous iron by food-reducing agents or brush border membrane enzymes before the body can use it, and then the divalent metal transporter needs to carry it into the enterocyte (Alaunyte et al., 2015). As non-haem iron absorption is dependent on the concentrations of enhancers and inhibitors, the availability of this type of iron varies widely, ranging from 2 to 20%. Since grain products provide 50%

of the total dietary iron in Western diets, non-haem iron makes up the majority of the iron that people consume (Alaunyte et al., 2015).

Iron metabolism and bioavailability:

Since the human body lacks a direct mechanism for excreting iron, the control of iron balance is dependent on both the individual's current iron status and the overall amount of iron-containing foods consumed. Internal homeostasis is responsible for maintaining this equilibrium (Alaunyte et al., 2015). Conversely, a person with low iron storage will absorb less iron than one with rich iron stores. Moreover, the total amount of iron as well as its type—hem or non-hem iron—as well as the levels of inhibitors and promoters—phytates, phenolic compounds, meat, fish, and shellfish, and calcium—that are present in the diet all affect how well the body absorbs iron from the diet as a whole (Alaunyte et al., 2015). Up to 60 μg/L, studies have shown an inverse connection between serum ferritin (sFer) concentration and iron absorption (Alaunyte et al., 2015). This shows that absorption drops to a level that covers daily iron losses, preventing additional iron storage. It was revealed that in persons with serum ferritin (sFer) content of 10 μg/L, 40% of the haem and non-haem iron was absorbed; at greater sFer concentrations, a decrease in iron absorption was noted, particularly in non-haem iron (Alaunyte et al., 2015). This suggests that overall iron status has a significant impact on iron absorption, independent of total dietary iron consumption. Reduced iron absorption occurs in an iron-depleted condition, whereas higher iron absorption from food occurs in an iron-replete state. Moreover, this implies that non-haem dietary iron becomes a significant source of absorbable iron in an iron-depleting state (Alaunyte et al., 2015).

Iron deficiency:

Three stages of iron deficiency progression are identified. First, a phenomenon known as iron storage depletion occurs when the iron stores in the reticuloendothelial cells of the liver, spleen, and bone marrow are reduced. This is shown by a decrease in serum ferritin. The second stage, erythropoiesis, is characterised by a decrease in iron transport and, consequently, a reduction in the iron supply to the cells. Low serum iron, an increase in total binding capacity, and a decline in transferrin saturation are the symptoms of this stage (Alaunyte et al., 2015). Iron-deficient non-anaemia or pre-anaemic "latent iron deficiency" are other terms used to describe the first two stages of iron shortage. Anaemia results from inadequate iron intake during the final stage of iron shortage, which impairs haemoglobin synthesis. Serum ferritin, or sFer, is widely accepted as the measure of iron reserves in healthy individuals (Alaunyte et al., 2015). Due to the close correlation between bone marrow iron reserves and plasma serum ferritin levels, low serum ferritin levels are indicative of latent iron insufficiency. Serum ferritin in mature females ranges physiologically from 15 to 200 µg/L. The lower limit of serum ferritin for diagnosis of latent iron shortage in female athletes is not well established because of this large inter-individual variation. Studies examining the iron status of a population of physically active females have varied lower limits, ranging from 12 to 35 µg/L. A value of 12 µg/L is generally regarded as indicative of an iron deficiency state in females, as this state is characterised by the absence of marrow iron stores and completely depleted iron stores (Alaunyte et al., 2015).

Systemic iron homeostasis:

Over the last ten years, a significant advancement in our understanding of iron metabolism has been made with the identification and characterisation of hepcidin, a peptide predominantly generated and secreted by hepatocytes (Buratti et al., 2015). The primary or only cellular iron exporter, ferroportin, is present on the cell plasma membrane, and hepcidin regulates its level of presence to maintain systemic iron homeostasis. Iron release into the plasma is inhibited by hepcidin binding, which causes ferroportin internalisation and destruction (primarily from enterocytes, macrophages, hepatocytes, and placental cells). As a result, there is less circulating iron and more iron is retained within cells (Buratti et al., 2015). Anaemia and siderosis are two examples of the many diseases linked to iron overload or deficiency that are brought on by any interference with the previously outlined mechanisms. In fact, the dysregulated duodenal iron absorption that characterises the majority of hereditary iron overload illnesses is caused by inadequate hepcidin expression in proportion to body iron reserves (Buratti et al., 2015). In order to stop more iron from entering the plasma, endoplasmic reticulum stress, inflammation, and iron overload all cause an increase in the production of hepatic hepcidin. On the other hand, hypoxia (a reaction that increases iron release and availability) or iron deficiency causes hepcidin to be down-regulated. In addition, hepcidin expression is regulated by metabolic switches, which are triggered by insulin, and hunger and suppressed in response to increased food levels through mTOR signalling (Buratti et al., 2015).

Cellular iron metabolism:

By adjusting the mRNA stability or the translation of important iron regulatory proteins in response to intracellular iron levels, iron regulatory proteins 1 and 2 (IRP1 and IRP2) serve as major regulators of iron balance in the cell (Buratti et al., 2015). IRP1 binds a Fe-S cluster to operate as a cytosolic aconitase, whereas IRP2 is redirected to proteasomal destruction mediated by the iron-sensing ubiquitin ligase FBXsL5, which suppresses both IRPs' ability to bind RNA in cells with sufficient iron storage. Both IRP1, which loses the Fe-S cluster, and IRP2, which is stabilised, bind to iron-responsive elements found in a range of transcripts implicated in iron intake, storage, utilisation, and export when there is an iron shortage (Buratti et al., 2015). More specifically, by stabilising the mRNA for the transferrin receptor (TfR1) and DMT1, strong IRP-binding activity increases iron uptake into the cell. Simultaneously, it inhibits the translation of the mRNAs encoding ferroportin and ferritin, so impeding the cell's ability to export iron and storage. In cells with low iron levels, this reaction increases iron availability. On the other hand, in iron-replete cells, reduced IRP activity leads to an increase in TfR1 mRNA degradation and permits the translation of mRNAs encoding ferroportin, ferritin, and other iron-consuming proteins. Thus, this sophisticated process prevents iron-mediated cell damage while simultaneously guaranteeing that adequate iron is accessible for cellular requirements, particularly for the mitochondria, which use the majority of cellular iron (Buratti et al., 2015).

Iron requirements and consumption in female athletes' diets:

The recommended dietary allowance (RDA) in the US is established at 18 mg of iron per day, whereas the reference nutrient intake (RNI) for adult females in the UK is 14.8 mg per day (Alaunyte et al., 2015). While more iron is advised for women who are pregnant or nursing, female athletes are not officially advised to consume more iron. While it is generally known that exercise causes the body to lose more minerals, including iron, there is little proof that this has a negative impact on the body's reserves. According to several authors, female endurance athletes especially distance runners need roughly a 70% boost in iron. The daily iron intake of 10 mg would be added to the 14.8 mg suggested by the UK if this recommendation were to be implemented (Alaunyte et al., 2015).

This suggests that low dietary iron intake may be a factor in the population's deficient iron status. This is consistent with a number of further cross-sectional research looking into how dietary iron sources affect female runners (Alaunyte et al., 2015). In a study by Nuviala et al., it was discovered that 68% of female runners (n = 25) consumed less iron than 15 mg per day. Similar iron intakes of 11.4 and 13.8 mg/day were noted by Hassapidou and Manstrantoni in middle distance runners during training and the competition season, respectively, while Snyder et al. revealed that their daily iron intake was approximately 14 mg, and their serum ferritin levels, which indicate low iron storage, ranged from 6 to 25 μg/L. The authors observed that runners who consumed vegetarianism had serum ferritin levels that were significantly ($P < 0.05$) lower (7.4 μg/L) than those of the meat-eating group (19.8 µg/L). This was primarily because vegetarian diets had lower iron bioavailability (0.66 mg of absorbable iron/day vs. 0.91 mg/day, $P < 0.05$), even though total iron intake was similar in both groups (Alaunyte et al., 2015).

Iron and athletic performance:

Both myoglobin and haemoglobin bind oxygen through the porphyrin ring of heme; myoglobin transfers oxygen from erythrocytes to muscle cells, while haemoglobin in red blood cells (RBCs) carries oxygen from the lungs to the exercising skeletal muscles (Hinton, 2014). Iron is essential to oxidative metabolism and thus is especially important for the endurance athlete whose athletic performance depends on a high aerobic capacity (for reviews specific to female athletes, see McClung (2012); DellaValle (2013)) (Hinton, 2014). The last stage of oxidative ATP generation, known as the electron transport chain, is dependent on nonheme iron-sulfur enzymes (NADH dehydrogenase, succinate dehydrogenase, and ubiquinone–cytochrome c reductase) as well as heme-containing cytochromes (a, a3, b, b5, c, and c1) (Beard and Tobin 2000). Therefore, low iron reduces the ability to make ATP through the anaerobic metabolism of glucose, which in turn lowers endurance capacity. As these iron-containing proteins are crucial for aerobic metabolism and oxygen utilisation, the presence or absence of iron in the body can influence an athlete's ability to perform at both their maximum and submaximal levels (Hinton, 2014).

Assessment of iron status:

To detect iron deficiency, including the most severe stage O iron-deficiency anaemia, blood tests are required (FNB 2001). Haemoglobin content in blood is one of the main markers for anaemia diagnosis. Hematocrit, or the percentage of red blood cells in blood volume, decreases with anaemia. African Americans have lower threshold haemoglobin and hematocrit values, while smokers and people who reside at altitude have higher threshold values. In the blood, iron is carried by a protein known as transferrin. With iron insufficiency, transferrin saturation—the percentage of transferrin transporting iron decreases. The appearance of the red blood cells (RBCs) can be utilised to differentiate iron deficiency anaemia from other nutritional anaemias (Hinton, 2014). Insufficient generation of haemoglobin is the outcome of iron insufficiency. The body produces new red blood cells quickly in an effort to compensate for the lower oxygen-carrying capacity. These RBCs appear tiny and extremely pale under a microscope due to their immaturity and lack of haemoglobin. Hypochromic microcytic anaemia is the name given to this kind of anaemia. Serum ferritin levels are measured to determine the extent of iron reserve depletion. Serum ferritin is utilised as an indication of iron storage because its concentration in blood is proportionate to that of ferritin stored in the liver (1 g ferritin/L is equal to 10 mg of stored iron) (Cook and Finch 1979). Regardless of iron reserves, the liver produces more ferritin during disease or inflammation since it is an acute-phase protein. Therefore, in order to prevent disguising decreased iron levels, ferritin should not be evaluated during illness or following a strenuous exercise session. The blood's level of sTfRs is utilised to determine the tissue iron status. (Punnonen et al. 1997) due to the fact that sTfR expression in cells rises with iron requirement (Skikne et al. 1990). Increased iron uptake occurs when the number of transferrin receptors on the cell surface rises, as in the case of skeletal muscle and bone marrow (Hinton, 2014).

Special considerations for iron-status assessment in athletic populations:

Iron insufficiency is more common in athletic populations, the International Olympic Committee recommends that assessments of iron status, including anaemia and iron reserves, be included in athletes' periodic health exams. Standardising measuring conditions is challenging when screening a large number of athletes, but it's crucial to understand that variables like the time of day, length of time since the last training session, illness, or injury may affect test validity and outcomes. True variations in markers of iron status can be obscured by abrupt changes in plasma volume. A seemingly low haematocrit or haemoglobin might emerge from endurance training's expansion of plasma volume, which may outpace or come before the adaptive increase in RBC quantity (Weight et al. 1991b). Known also as "sports anaemia" or "dilutional anaemia," this condition has no negative effects on performance. Elevated hemolysis is indicated by lower serum haptoglobin, hematuria, and hemoglobinuria, as well as indications of anaemia, iron storage, and tissue iron status (Hinton, 2014).

Consequences of poor iron status:

When iron deficiency occurs, several organs exhibit morphologic, physiological, and biochemical alterations that are connected to the turnover of vital iron-containing proteins. This might occasionally happen even prior to a noticeable drop in haemoglobin concentration (Beard & Tobin, 2000). Reduced mitochondrial electron transport, neurotransmitter production, protein synthesis, organogenesis, and other metabolic activities are linked to iron shortage. Glossitis, angular stomatitis, koilonychia (spoon nails), blue sclera, esophageal webbing (Plummer-Vinson syndrome), and anaemia are the overt physical signs of chronic iron deficiency. People with iron deficiency frequently have behavioural abnormalities including pica, which is the abnormal eating of dirt (geophagia) and ice (pagophagia). However, a biological basis for these behavioural abnormalities is unknown It's also critical to distinguish between the possible effects of exercise on iron status and whether those effects are harmful to an athlete's health or ability to perform well (Beard & Tobin, 2000).

Exercise and iron status of non-athletes:

There isn't much research on how exercise training or fitness-type exercise affects those who were previously inactive. It follows that the lack of conclusive results is not surprising. It is challenging to make comparisons between research because of the wide variations in the subjects' ages, types of exercise, and exercise durations (Evans, 1992). Kilbom et al. observed that bicycle ergometer training at 70% maximal aerobic power resulted in minor drops in haemoglobin and hematocrit levels as well as a 25% fall in serum iron in adult women aged 19–64. Still, Wirth et al. (13 when data were compared with values from control participants, previously untrained college women on a training schedule of three 20–25 minutes cycling rides per week for 10 weeks showed no significant change in haemoglobin, hematocrit, or serum (Evans, 1992).

Since achieving optimal performance is a constant goal for athletes, both for their team and for themselves, a significant portion of their time is spent improving performance through off-season conditioning, sport-specific training, and dietary adjustments (1). As a result, it is not unusual to find many athletes training or competing nonstop throughout the year (Gropper et al., 2006).

Haematological and biochemical marker testing is a common part of sports medicine and science care, and it has a significant impact on athletes' growth and performance. There is evidence in elite sports that low iron levels are strongly associated with decreased well-being among athletes as well as low aerobic ability. Ferritin functions as an additional haematological measure, while serum iron merely represents the circulating iron (Arabi & Piert, 2010). Haematological parameters like the hematocrit, haemoglobin, and red blood cell count are also necessary to provide a comprehensive and precise picture of the blood's oxygen-transporting capacity. It may take a few weeks for other measurements, such as the mean cell volume of red blood cells (MCV), to significantly alter and indicate a shortage (Arabi & Piert, 2010). It is therefore advised to measure reticulocytes and the factors they produce in order to identify shortages early on. Since reticulocyte haemoglobin (CHr or RetHe) is a marker of the early phases of iron demand in erythropoiesis before the onset of anaemia, it has been used to monitor iron status (Arabi & Piert, 2010).

Iron deficiency anaemia (IDA) is a global health issue that causes a variety of symptoms, such as weariness and weakness. The third, last, and most severe stage of iron insufficiency is known as IDA.21:38 All together, the two early stages—which precede IDA—are known as iron deficiency nonanemic (IDNA). A decrease in sFer due to the depletion of total body iron stores while maintaining normal levels of other iron indices and normal haemoglobin is indicative of nonanemic, the initial and least severe stage of iron shortage (Arabi & Piert, 2010). Low sFer, low serum iron or decreased transferrin saturation, and enhanced total iron binding capacity (TIBC) are characteristics of the second stage, which is also non-anaemia. The body can no longer meet the demands of haemoglobin synthesis once iron reserves and transport iron have been sufficiently depleted, leading to the third and final stage. It is now standard procedure for endurance athletes of all levels—not just those with a history of IDA—to monitor their sFer concentration (Rubeor et al., 2018).

DISCUSSION

Iron deficiency: prevention and treatment by consuming a diet high in iron. The primary goal of iron deficiency prevention in endurance athletes should be to increase the intake of foods high in iron and foods that improve iron absorption through diet. The average American woman finds it challenging to get the required daily intake of 18 mg of iron by diet alone because the food supply in the country contains only 6 mg of iron per 1000 calories. In fact, a woman would need to eat 3000 calories in order to receive 18 mg of iron, which is more energy than many female athletes actually need or consume Therefore, it is especially crucial to consume foods high in iron, such as meat, grains and breakfast cereals enriched with iron, dried fruit, nuts, and soy nuts. A considerable portion of "energy bars" may be included in one's overall iron intake because they are fortified with iron as well. Information on choosing foods with high iron bioavailability should be given to athletes. Vegetarian athletes must take extra care to ensure they ingest adequate iron because the bioavailability of plant-based iron (nonheme iron) is much lower than that of animal-based iron (heme and nonheme iron)(Hinton, 2014).

Iron supplementation

When an athlete has serum ferritin levels less than 20 ng/mL or iron-deficiency anaemia (Pitsis et al. 2004). Research on athletes with nonanemic iron insufficiency, both male (Nachtigall et al., 1996) and female (Hinton and Sinclair, 2007; DellaValle and Haas, 2014), has shown that consuming 20–100 mg of elemental iron daily for 6–12 weeks effectively raises serum ferritin concentrations. The Centres for Disease Control (CDC, 1998) recommend an oral intake of 60–120 mg of elemental iron per day, divided into two doses of 50–60 mg. This amount of iron is present in several over-the-counter iron-only supplements per dose. For instance, 65 mg of elemental iron is contained in a 325 mg ferrous sulphate pill. Based on gastrointestinal symptoms linked to supplemental iron ingestion, the upper intake level (UL) for iron is 45 mg/day (FNB 2001). The UL is applicable to individuals with normal iron status, and doctors frequently administer iron at dosages greater than 45 mg per day to treat anaemia, in accordance with the CDC's advice. After four weeks of supplementation, individuals receiving supplemental iron should undergo another test. If treatment results in a rise in haemoglobin, the supplementation should be continued for a further two to three months in order to replenish iron storage (Hinton, 2014).

Iron overload

For people with normal iron status, the UL for iron is 45 mg/day (FNB 2001). Serum ferritin levels of more than 200 ng/mL, or iron excess, is a serious complication of unneeded or improperly regulated iron supplementation. According to Mettler and Zimmermann (2010), iron excess affected 15% of male marathon runners and 5% of female runners (Hinton, 2014). Body iron stays unbound to proteins when it surpasses ferritin's and transferrin's respective storage and transport capacities. Lipid peroxidation and the generation of free radicals are brought on by this free iron, and these processes harm the liver, kidneys, heart, and central nervous system. Furthermore, the risk of iron overload may be elevated by genetic diseases that impact iron metabolism (Hinton, 2014). People who have the autosomal recessive disorder hereditary hemochromatosis absorb iron excessively and have poor storage, which damages several organs. Iron overload is also more likely in cases of thalassemia or sideroblastic anaemia due to increased iron absorption resulting from higher erythropoiesis. Iron supplementation to address a deficit should be supervised by a healthcare provider due to the risk of iron overload. Before starting an additional regimen, athletes who exhibit signs that are indicative of an iron deficiency—such as unusual fatigue or performance declines, cold sensitivity, or recurrent illnesses—should have their iron status evaluated. Following the diagnosis of iron deficiency and the start of supplementation (e.g., 65 mg elemental iron/day), the athlete's iron status should be checked on a frequent basis (e.g., every 4-6 weeks).) (Hinton, 2014).

Iron deficiency in sports therapy: Nutrition

Adjusting the dietary iron intake is the first step in treating iron deficiency. Iron found in meat and free iron in the form of Fe2+ or Fe3+ are both present in nutrition. Studies on oral absorption reveal that haemoglobin is far more readily absorbed than free iron (Clénin et al., 2016). Fe2+ uptake is preferable to Fe3+ uptake for the latter. Together with free iron, meat, liver, poultry, and fish also contain haemoglobin. Only free iron is present in a vegetarian diet. Iron's bioavailability varies greatly and is mostly dependent on actual iron reserves with values ranging from 5% to 15%. A notable increase in iron bioavailability of up to 35% is found in the case of iron insufficiency Furthermore, enhancers and inhibitors are among the nutritional factors that affect the digestive tract's absorption of iron (Hinton, 2014).. Vitamin C, peptides from partially digested muscle tissue, fermented foods, and organic acids like citrate or malate are substances that increase the absorption of iron. Phytates, oxalates, polyphenols (found in black tea and coffee), peptides derived from partially digested vegetable proteins, and calcium are substances that limit the uptake of iron (Hinton, 2014).. 14 milligrams of nutrients should be consumed daily. A sufficient calorie intake is generally advised for the best dietary iron intake in sports, particularly for athletes with low body mass index who are more likely to experience iron insufficiency [95]. It is still up for debate whether low-energy-related catabolism affects hepcidin control and inhibits iron absorption. Since meat, poultry, and fish are the primary sources of nutritional iron intake, it is generally advised to consume them on a regular basis—at least five times per week. It is advised to eat wholemeal products in addition to legumes and green vegetables every day (Hinton, 2014).

Oral iron

Oral iron treatment is typically used in conjunction with dietary guidance. The iron (Fe2+ or Fe3+), their complex-forming substrate, and their galenical shape are different in oral preparations. Innovative products mix vitamin C and iron (Hinton, 2014). Three dosages of oral iron were examined in a dose-finding research conducted by Rimon et al. in elderly individuals with IDA: 15 mg, 50 mg, and 150 mg. They were able to demonstrate that iron supplementation at the RDA level—which is 15 mg of elemental iron—already significantly improved the iron status. The doses of 50 mg and 150 mg of elemental iron in these elderly anaemic individuals did not show any further benefit, but they did have a notably higher number of side effects. The suggested quantity of oral iron should not be excessive because there is additional evidence that oral iron loading raises circulation hepcidin (Hinton, 2014).

Intravenous iron

IV therapy should be taken into consideration when oral medication fails or an emergency restoration is required. Iron saccharose and ferric carboxymaltose are currently offered in two separate formulations in Switzerland. The degree of the iron shortage determines the dosage. 500–100 mg of Fe-carboxymaltose or 200 mg of Fe-saccharose can be given in a single application. The prompt treatment of iron deficiency and replenishment of depleted iron reserves is the primary benefit of intravenous (IV) therapy. In general, patients follow their iron supplement regimen well. Side effects can include a brief loss of taste, headache, light headedness, myalgia, and fever, but extremely rare cases of severe adverse reactions, including hypotonic and anaphylactoid responses, tachycardia and arrhythmia, dyspnea, and bronchospasm (Hinton, 2014).

CONCLUSION

Dietary changes and supplementation are key components in the prevention and treatment of iron deficiency in endurance athletes. Increasing iron intake and improving iron absorption through food are the objectives. Iron-rich foods such as meat, grains, enriched cereals, dried fruit, nuts, soy nuts, and fortified energy bars become essential. Because plant-based iron has a lower bioavailability than animal-based iron, it is important to educate athletes about selecting foods with higher bioavailability of iron, especially vegetarian athletes. Iron supplementation becomes necessary when serum ferritin levels fall below 20 ng/mL or when iron-deficiency anaemia manifests. Research on iron deficiency in athletes has demonstrated that increasing blood ferritin concentrations by 20–100 mg of elemental iron per day for 6–12 weeks is an effective treatment. The Centres for Disease Control (CDC) recommends an oral consumption of 60–120 mg of elemental iron per day, with a daily maximum intake of 45 mg. On the other hand, too much iron can result in iron overload, which can harm the kidneys, heart, liver, and nervous system. It is critical to comprehend the hazards associated with iron excess, even for those with normal iron status. The risk of iron overload can be increased by genetic illnesses that affect iron metabolism, such as hereditary hemochromatosis. Therefore, prior to starting supplements, athletes who exhibit symptoms of iron insufficiency should have their iron status evaluated. Nutritional changes are advised, with a focus on including foods high in vitamin C, such as meat, liver, poultry, and fish. Iron's bioavailability, however, fluctuates, and dietary enhancers and inhibitors have a big impact on how well iron is absorbed. Oral iron treatment involves different formulations and dosages and is frequently used in conjunction with dietary advice. Research indicates that high oral iron loading increases the level of hepcidin in the blood, which controls the absorption of iron. When oral medication is ineffective or an immediate iron replenishment is required, intravenous (IV) therapy may be explored. Although IV therapy quickly raises iron levels, there is a chance of temporary taste loss, headaches, and fever as adverse effects.

A comprehensive strategy that balances dietary modifications, oral supplementation, and, in extreme situations, intravenous therapy is required for the prevention and treatment of iron deficiency in athletes. In order to manage iron shortage and prevent issues associated with iron overload, it is essential to understand each person's unique dietary needs and to regularly check their iron status.

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